ExoHabitAI – ML Model Training (Milestone-2)

1. Objective

The objective of this phase is to select, train, evaluate, and finalize the best-performing machine learning model for predicting the habitability potential of exoplanets using preprocessed planetary and stellar data.

The model must classify exoplanets as:

- **1 → Potentially Habitable**
- **0 → Non-Habitable**

Additionally, the model should generate a probability score representing habitability likelihood.

2. Problem Formulation

Learning Type

Supervised Machine Learning

Prediction Type

Binary Classification

Input

Cleaned and engineered numerical and encoded categorical features derived from exoplanet and stellar data.

Output

- Habitability class (0 or 1)
- Habitability probability score

3. Dataset Preparation

Input File

`exohabit_ml.csv`

Steps Performed

1. Loaded preprocessed dataset.
2. Converted multi-class habitability into binary classification:
   - 0 → Non-habitable
   - 1 & 2 → Habitable
3. Removed leakage-related features (e.g., habitability_index and derived scoring features).
4. Separated:
   - X → Feature matrix
   - y → Target variable
5. Performed train-test split:
   - 80% training
   - 20% testing
   - Stratified split to maintain class balance
   - `random_state=42` for reproducibility

4. Baseline Model – Logistic Regression

Logistic Regression was used as a baseline model.

Performance (Test Set)

- Accuracy: 96.9%
- Recall (Habitable class): 0.92
- F1-score: 0.94
- ROC-AUC: 0.992

Interpretation

Logistic Regression performed well but showed lower recall compared to advanced models. Since planetary habitability involves non-linear interactions, a linear model may not fully capture complex relationships.

5. Advanced Model 1 – Random Forest

Random Forest was trained to capture non-linear relationships and feature interactions.

Performance (Test Set)

- Accuracy: 98.49%
- Recall (Habitable class): 0.97
- F1-score: 0.97
- ROC-AUC: 0.9991

Interpretation

Random Forest significantly improved recall and F1-score compared to Logistic Regression. It effectively handled non-linear patterns in planetary features.

6. Advanced Model 2 – XGBoost (Final Model)

XGBoost was trained as a boosting-based ensemble model for optimized performance on structured data.

Performance (Test Set)

- Accuracy: 99.70%
- Recall (Habitable class): 0.99
- F1-score: 0.99
- ROC-AUC: 0.99995

Training vs Testing Accuracy

- Training Accuracy: 100%
- Testing Accuracy: 99.70%

This indicates minimal overfitting and strong generalization.

7. Cross-Validation

5-fold cross-validation was performed on the XGBoost model.

F1 Scores:

0.9725, 0.9919, 0.9911, 0.9930, 0.9897

Average F1-score: **0.9877**

Interpretation

The small variance across folds indicates:

- High model stability

- Strong generalization capability
- Robust performance across different splits

8. Hyperparameter Tuning

RandomizedSearchCV was used to tune:

- n_estimators
- max_depth
- learning_rate
- subsample
- colsample_bytree

The tuned XGBoost model achieved improved F1-score and maintained high stability.

This confirms that boosting methods effectively optimize classification boundaries for habitability prediction.

9. Model Comparison

Model	Accuracy	Recall	F1-score	ROC-AUC
Logistic Regression	96.9%	0.92	0.94	0.992
Random Forest	98.49%	0.97	0.97	0.9991
XGBoost	99.70%	0.99	0.99	0.99995

Final Model Selection

XGBoost was selected because:

- Highest recall (0.99)
- Highest F1-score (0.99)
- Highest ROC-AUC
- Strong cross-validation stability
- Ability to model complex non-linear feature interactions

10. Feature Importance & Interpretability

Top contributing features:

1. Planetary radius (pl_rade)
2. Equilibrium temperature (pl_eqt)
3. Orbital distance (pl_orbsmax)

4. Orbital period (pl_orbper)
5. Stellar temperature (st_teff)

Scientific Interpretation

- Planet radius determines whether a planet is rocky or gaseous.
- Equilibrium temperature affects liquid water stability.
- Orbital distance influences habitable zone positioning.
- Stellar properties determine radiation environment.

These findings align with astrophysical habitability principles.

11. ROC Curve Analysis

The ROC curve shows near-perfect separation between habitable and non-habitable classes.

AUC ≈ 1.0

This indicates extremely strong class discrimination.

12. Habitability Scoring & Ranking

The final XGBoost model was used to:

- Generate habitability probability for each exoplanet.
- Rank planets from most to least habitable.
- Save results in `habitability_ranked.csv`.

13. Final Conclusion

The XGBoost model effectively predicts exoplanet habitability using planetary and stellar features. It achieved high recall and F1-score while maintaining strong generalization across cross-validation folds.

Feature importance analysis confirms alignment with astrophysical theory, validating the scientific reliability of the model.

Therefore, XGBoost was selected as the final model for ExoHabitAI.